

Headline metrics — std log-IV space (overall, n_test=1007) (pred_len=5)

display	RMSE [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
persistence	0.2511 [0.2179, 0.2880]	0.0631	0.1648	— (reference)	1.000	—	—
dlinear	0.2561 [0.2171, 0.2974]	0.0656	0.1668	-0.0401 [-0.1098, +0.0358]	0.942	96,000	0.0626
var	0.2678 [0.2233, 0.3145]	0.0717	0.1692	-0.1374 [-0.2419, -0.0060]	0.934	—	—
tucker_dlinear	0.2697 [0.2302, 0.3120]	0.0727	0.1785	-0.1533 [-0.2265, -0.0721]	0.924	11,566	0.0618
gwn	0.2700 [0.2247, 0.3190]	0.0729	0.1756	-0.1559 [-0.2795, -0.0229]	0.929	10,841	0.0637
patchtst	0.3019 [0.2494, 0.3597]	0.0912	0.2001	-0.4455 [-0.6977, -0.2158]	0.836	3,318	0.0617
hot/kronecker_sum	0.3119 [0.2547, 0.3783]	0.0973	0.1981	-0.5432 [-0.8968, -0.2376]	0.921	166,725	0.0672
hot/kronecker_product	0.3179 [0.2496, 0.3943]	0.1011	0.1953	-0.6026 [-1.0853, -0.2039]	0.921	83,909	0.0645

Headline metrics — COVID (n_test=254) (pred_len=5)

display	RMSE [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
persistence	0.3580 [0.2662, 0.4474]	0.1282	0.2216	— (reference)	1.016	—	—
dlinear	0.3651 [0.2564, 0.4678]	0.1333	0.2199	-0.0398 [-0.1555, +0.1206]	0.964	96,000	0.0626
tucker_dlinear	0.3859 [0.2795, 0.4883]	0.1489	0.2427	-0.1619 [-0.2756, -0.0272]	0.956	11,566	0.0618
gwn	0.3945 [0.2671, 0.5124]	0.1557	0.2358	-0.2145 [-0.4203, +0.0655]	0.951	10,841	0.0637
var	0.4028 [0.2890, 0.5157]	0.1623	0.2418	-0.2661 [-0.4639, -0.0447]	0.963	—	—
patchtst	0.4488 [0.3064, 0.5860]	0.2014	0.2835	-0.5718 [-1.0056, -0.1338]	0.851	3,318	0.0617
hot/kronecker_sum	0.4946 [0.3446, 0.6429]	0.2446	0.3103	-0.9085 [-1.5551, -0.3563]	0.947	166,725	0.0672
hot/kronecker_product	0.5157 [0.3307, 0.6802]	0.2659	0.2996	-1.0750 [-1.9845, -0.2957]	0.893	83,909	0.0645

Headline metrics — Reflation calm (n_test=252) (pred_len=5)

display	RMSE [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
var	0.2028 [0.1716, 0.2344]	0.0411	0.1433	+0.1401 [+0.0588, +0.2216]	0.908	—	—
dlinear	0.2114 [0.1782, 0.2422]	0.0447	0.1489	+0.0661 [-0.0183, +0.1624]	0.928	96,000	0.0626
gwn	0.2140 [0.1803, 0.2470]	0.0458	0.1511	+0.0431 [-0.0620, +0.1606]	0.902	10,841	0.0637
tucker_dlinear	0.2153 [0.1819, 0.2485]	0.0464	0.1511	+0.0309 [-0.0937, +0.1582]	0.874	11,566	0.0618
patchtst	0.2177 [0.1808, 0.2539]	0.0474	0.1515	+0.0097 [-0.1410, +0.1710]	0.834	3,318	0.0617
persistence	0.2187 [0.1855, 0.2503]	0.0478	0.1523	— (reference)	1.002	—	—
hot/kronecker_product	0.2191 [0.1896, 0.2507]	0.0480	0.1597	-0.0029 [-0.1077, +0.0823]	0.885	83,909	0.0645
hot/kronecker_sum	0.2260 [0.1911, 0.2611]	0.0511	0.1595	-0.0676 [-0.2058, +0.0703]	0.860	166,725	0.0672

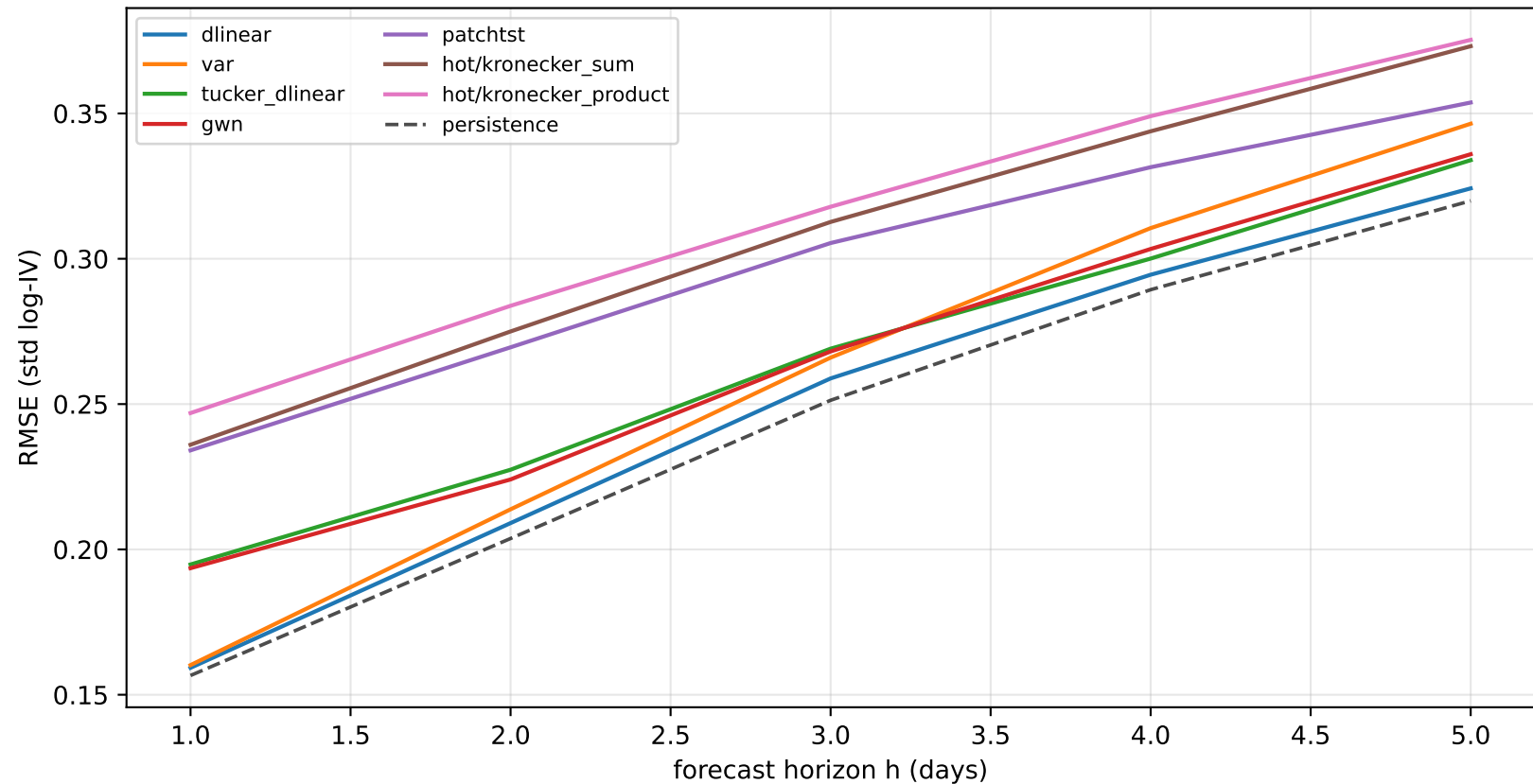
Headline metrics — Bear 2022 (n_test=251) (pred_len=5)

display	RMSE [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
persistence	0.2141 [0.1864, 0.2451]	0.0458	0.1557	— (reference)	1.031	—	—
var	0.2275 [0.1947, 0.2654]	0.0518	0.1616	-0.1298 [-0.3003, +0.0390]	0.949	—	—
dlinear	0.2291 [0.1982, 0.2632]	0.0525	0.1644	-0.1450 [-0.2593, -0.0295]	0.978	96,000	0.0626
hot/kronecker_product	0.2349 [0.2053, 0.2672]	0.0552	0.1774	-0.2045 [-0.4010, -0.0405]	1.013	83,909	0.0645
gwn	0.2395 [0.2095, 0.2739]	0.0574	0.1781	-0.2518 [-0.3826, -0.1253]	0.992	10,841	0.0637
hot/kronecker_sum	0.2464 [0.2147, 0.2848]	0.0607	0.1829	-0.3246 [-0.5545, -0.1157]	0.887	166,725	0.0672
tucker_dlinear	0.2493 [0.2185, 0.2854]	0.0621	0.1832	-0.3559 [-0.5342, -0.1888]	0.978	11,566	0.0618
patchtst	0.2911 [0.2591, 0.3280]	0.0848	0.2336	-0.8498 [-1.2458, -0.5292]	0.825	3,318	0.0617

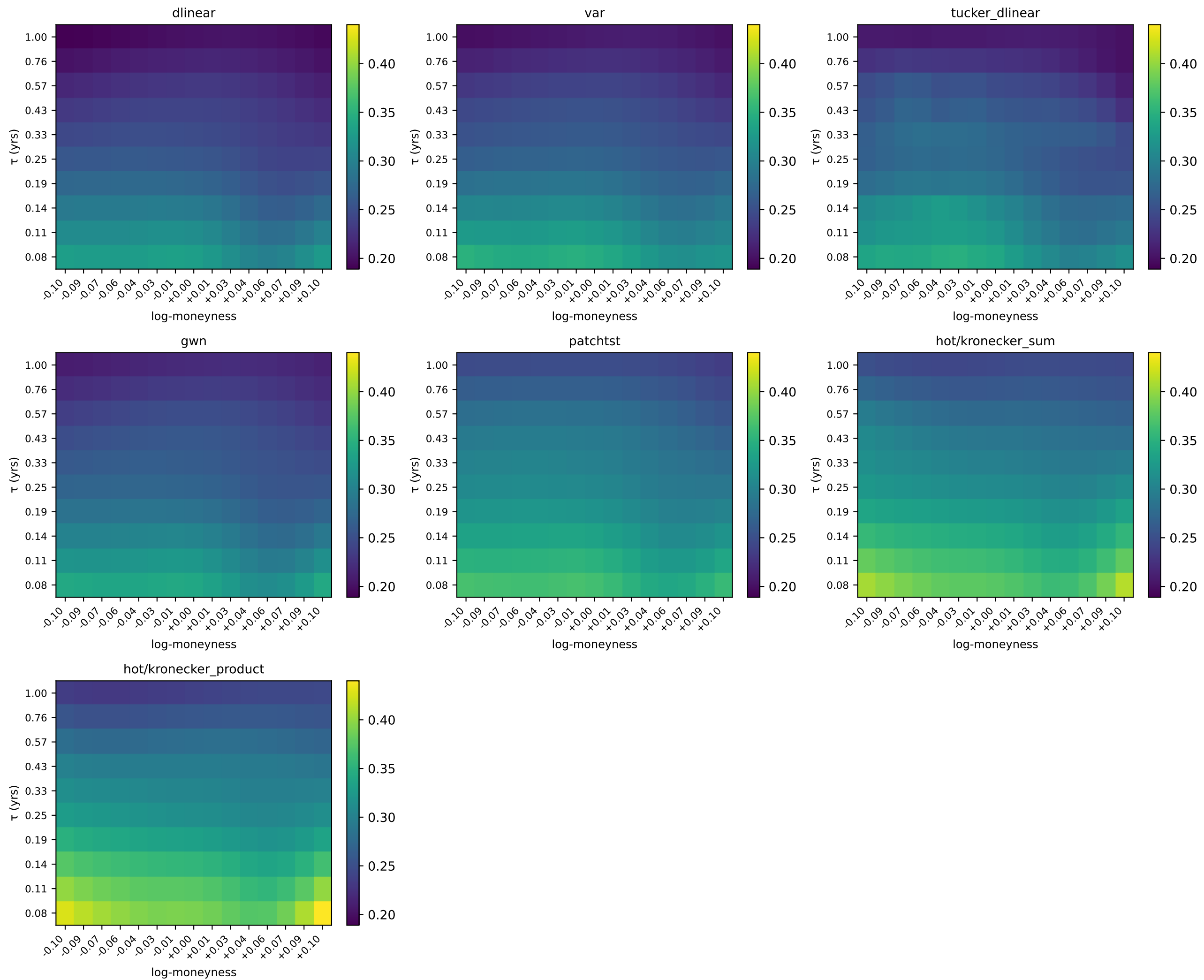
Headline metrics — Normalisation (n_test=250) (pred_len=5)

display	RMSE [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
persistence	0.1719 [0.1529, 0.1944]	0.0295	0.1287	— (reference)	1.020	—	—
patchtst	0.1721 [0.1531, 0.1935]	0.0296	0.1309	-0.0027 [-0.1136, +0.1103]	0.883	3,318	0.0617
var	0.1749 [0.1549, 0.1985]	0.0306	0.1293	-0.0349 [-0.1484, +0.0683]	0.971	—	—
hot/kronecker_sum	0.1761 [0.1608, 0.1941]	0.0310	0.1383	-0.0497 [-0.1952, +0.0651]	0.925	166,725	0.0672
dlinear	0.1762 [0.1567, 0.1998]	0.0310	0.1334	-0.0509 [-0.1441, +0.0318]	0.963	96,000	0.0626
gwn	0.1781 [0.1609, 0.1983]	0.0317	0.1366	-0.0734 [-0.2052, +0.0398]	0.928	10,841	0.0637
tucker_dlinear	0.1803 [0.1590, 0.2049]	0.0325	0.1360	-0.1004 [-0.2499, +0.0377]	0.895	11,566	0.0618
hot/kronecker_product	0.1818 [0.1665, 0.2000]	0.0331	0.1433	-0.1193 [-0.2965, +0.0068]	1.026	83,909	0.0645

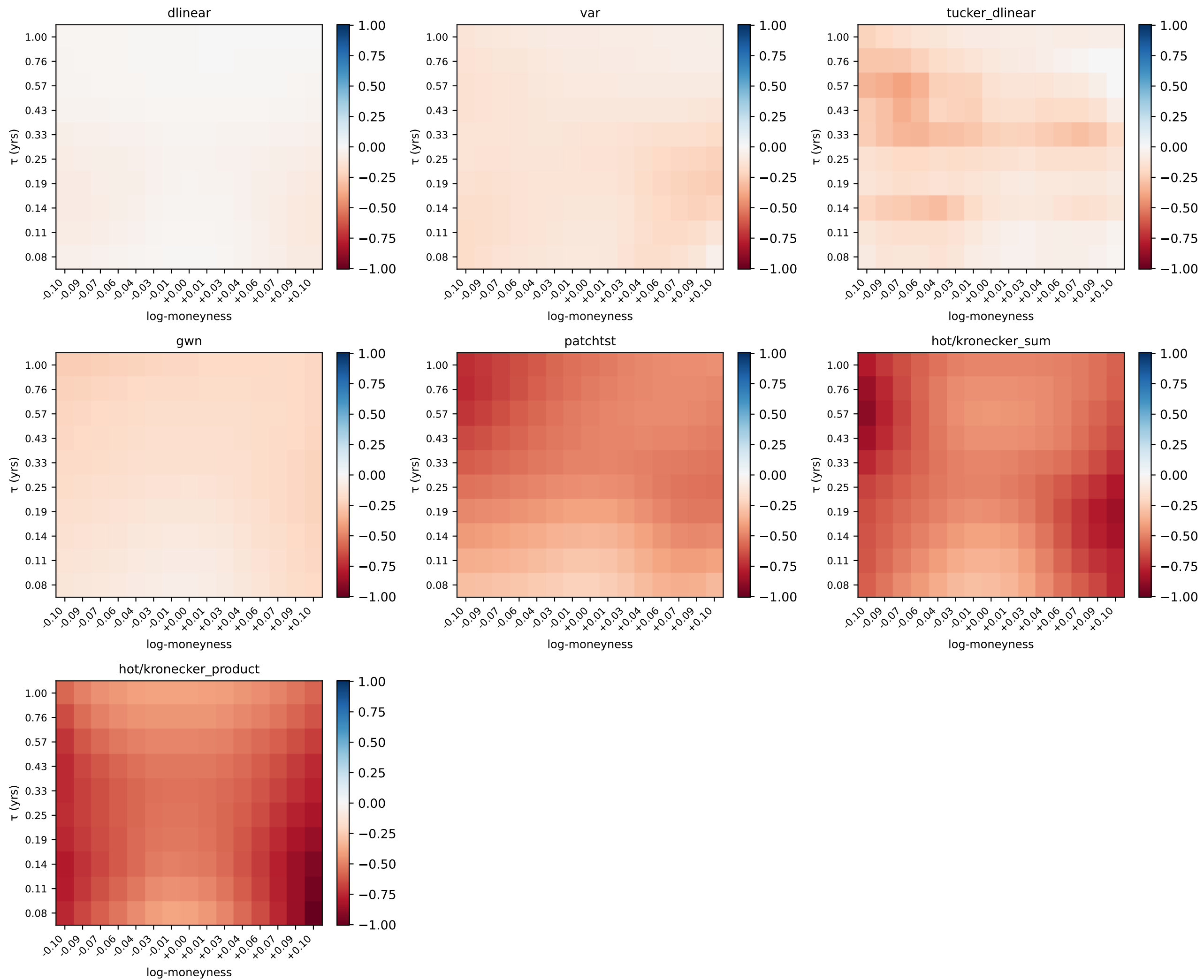
RMSE vs forecast horizon



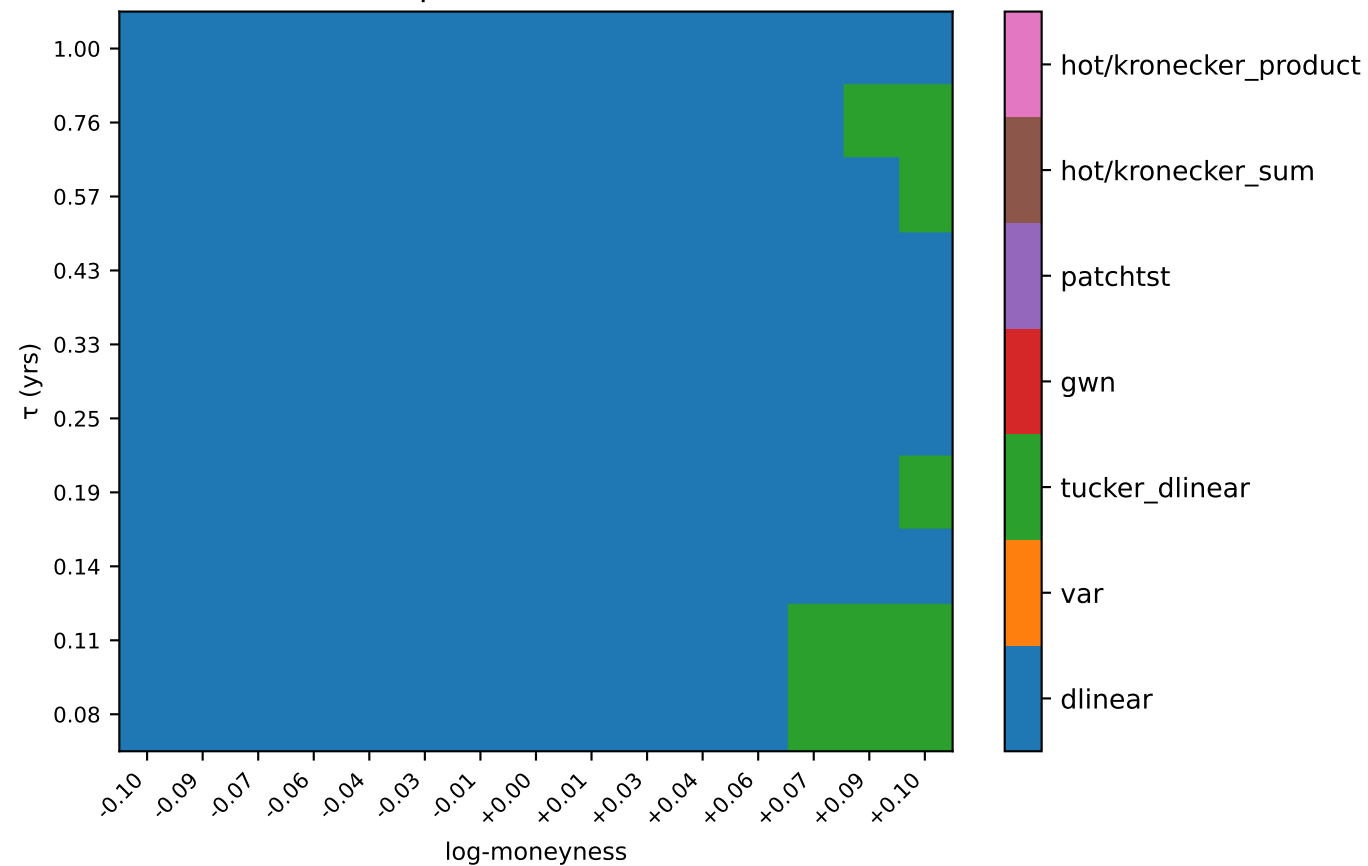
Per-cell RMSE (std log-IV) — each panel is one model



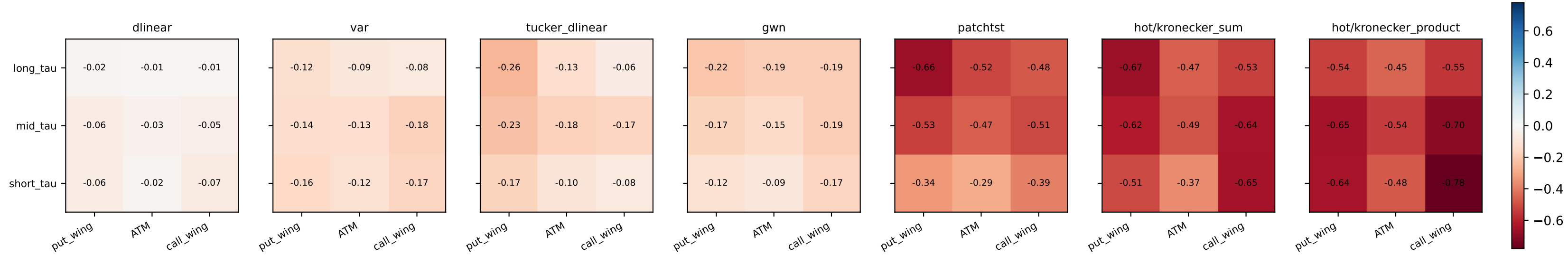
R^2 vs persistence per cell (red = worse than persistence)



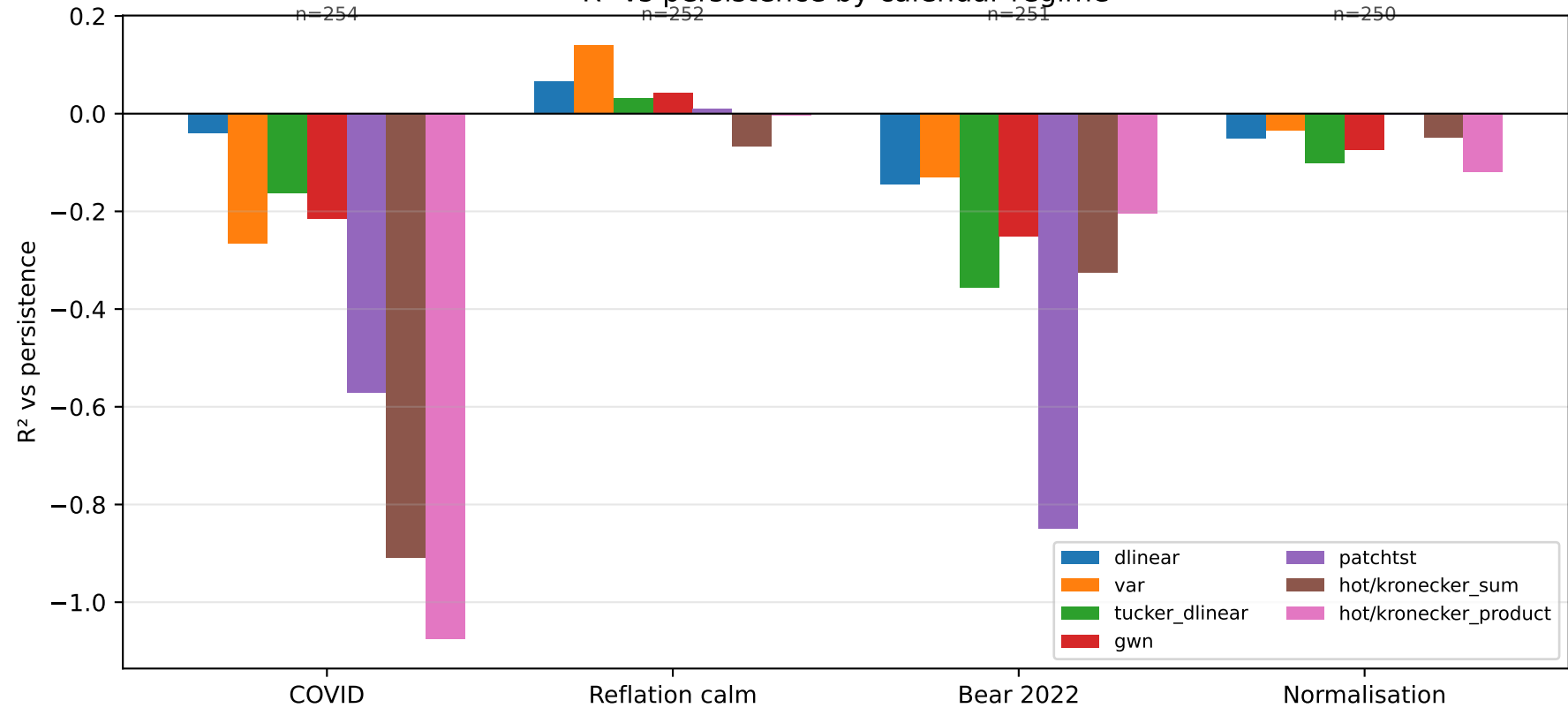
Winner per cell (lowest RMSE)



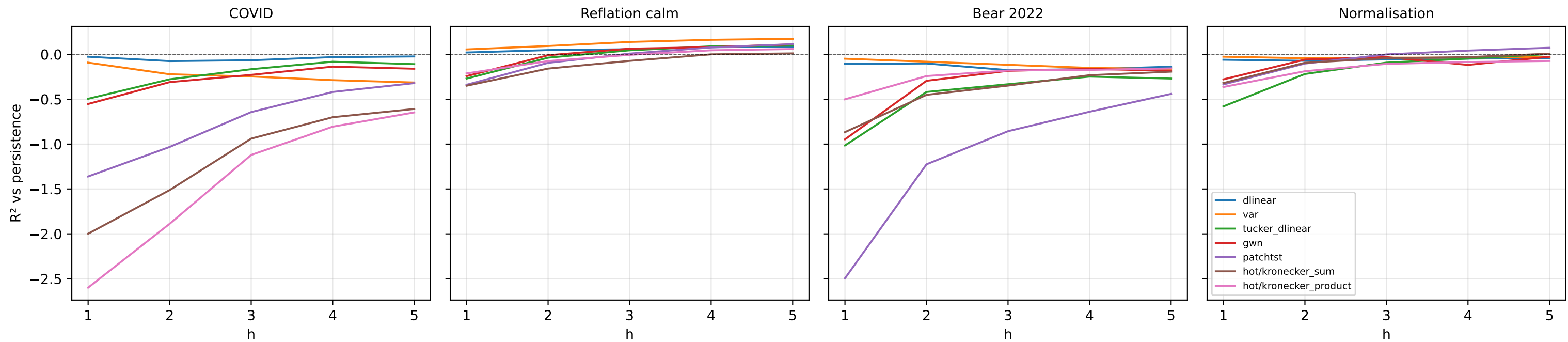
R² vs persistence by surface zone

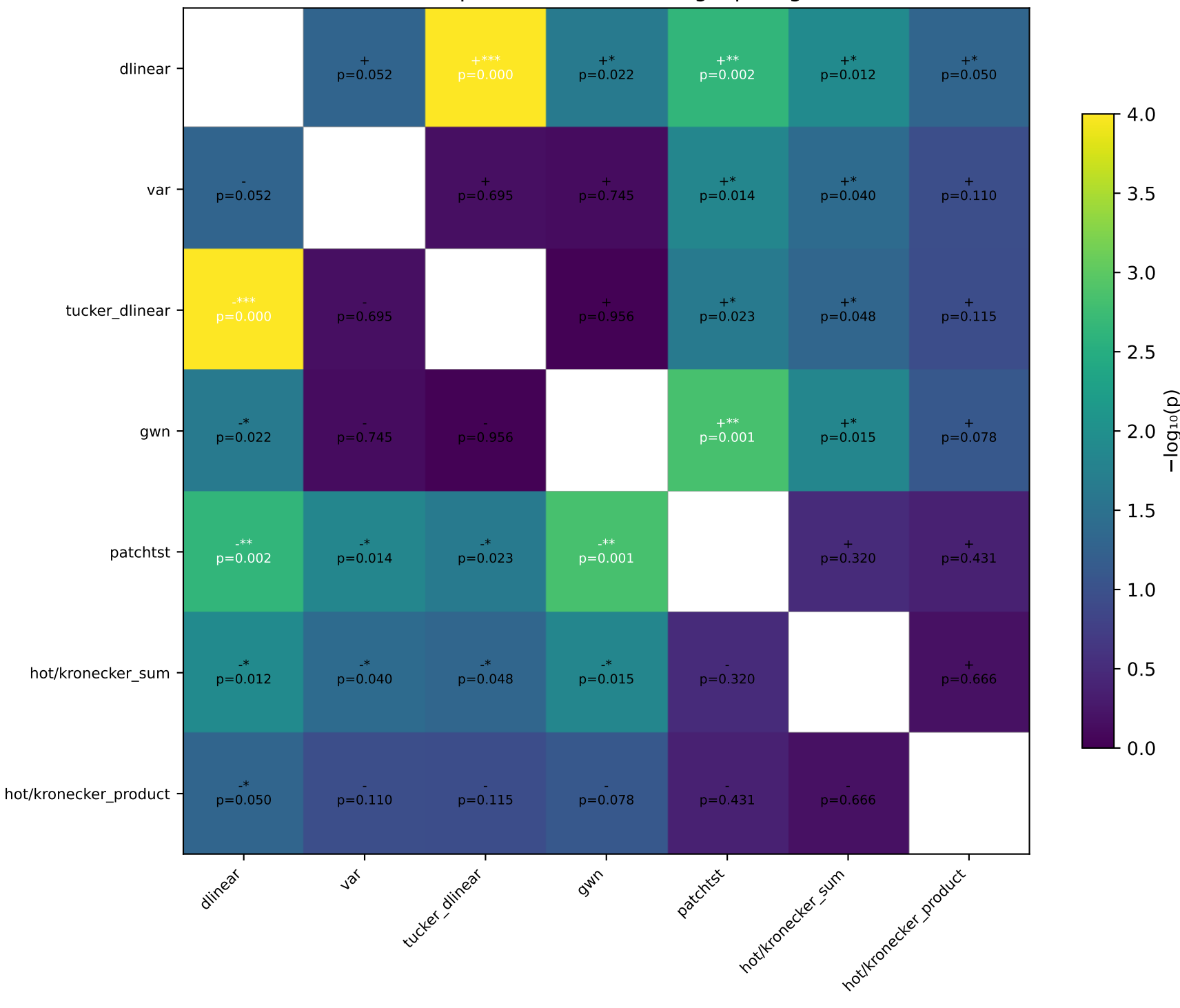


R² vs persistence by calendar regime

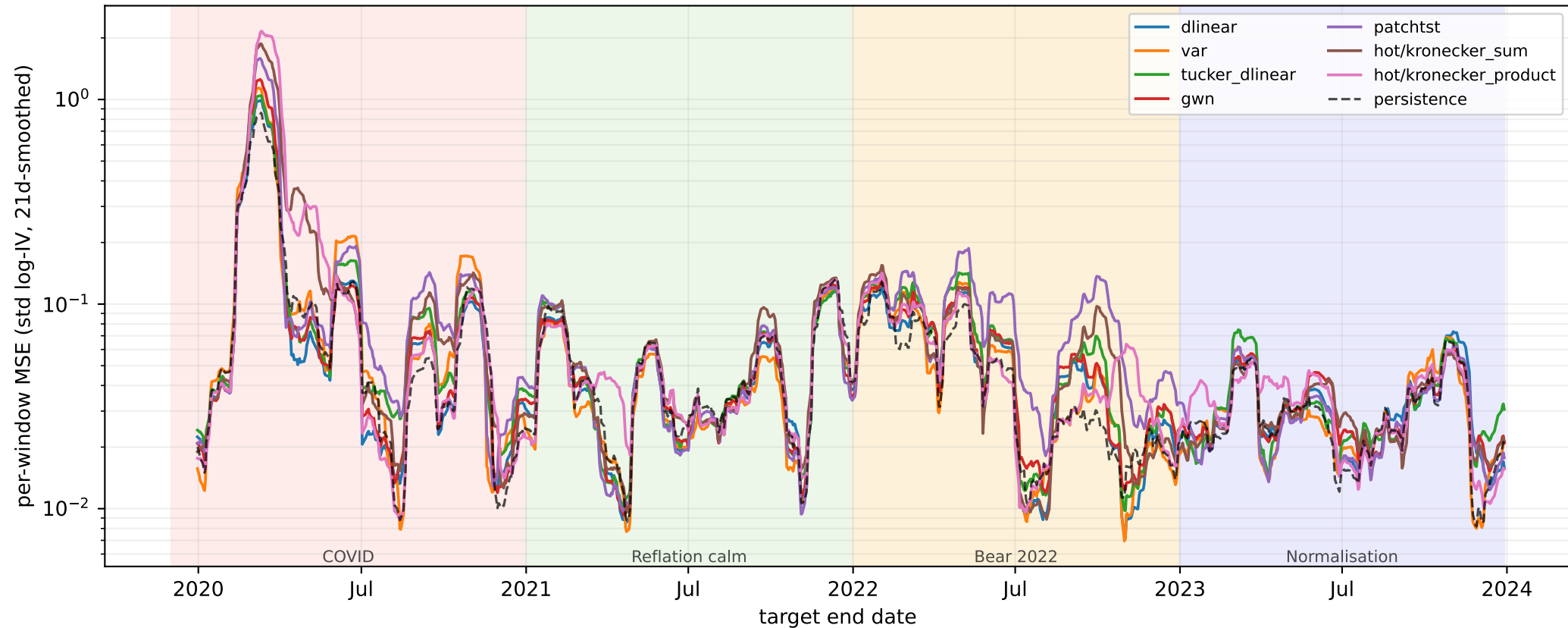


R^2 vs persistence by horizon, faceted by regime

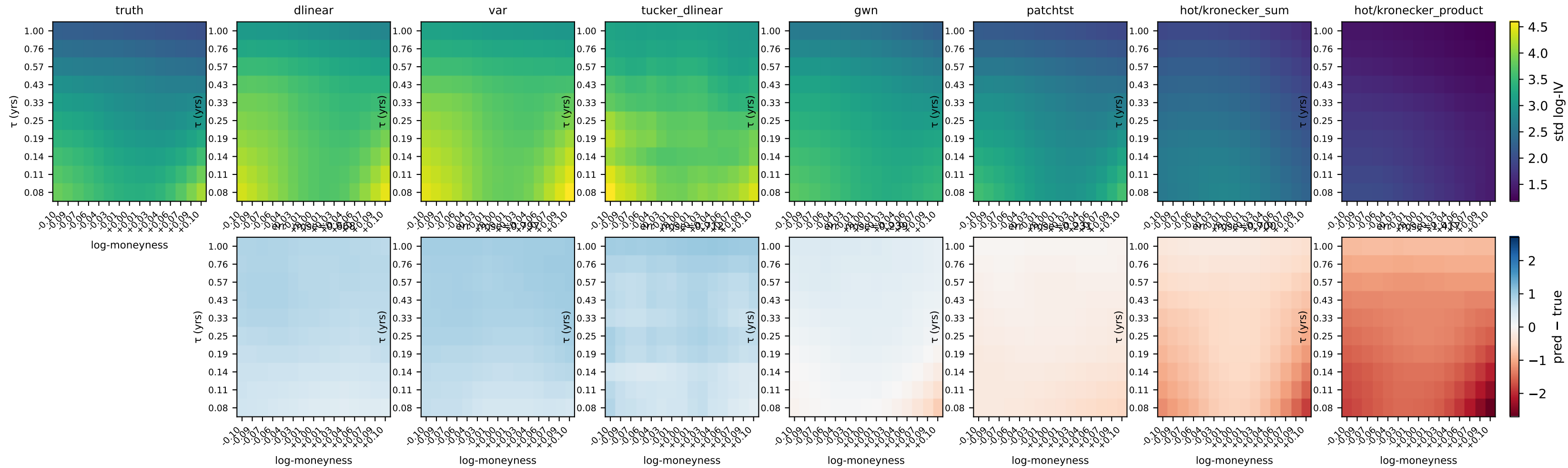


Diebold-Mariano pairwise — color: $-\log_{10}(p)$, sign: row vs col

Per-window MSE over time, regime-shaded



Example forecast — covid: target 2020-03-23 (h=5)



Example forecast — bear: target 2022-06-15 (h=5)

